

Senbao Lu

100 Institute Rd., Worcester MA • +1(669)246-2445
slu3@wpi.edu • ORCID: [0000-0001-9612-9703](https://orcid.org/0000-0001-9612-9703)

EDUCATION BACKGROUND

Worcester Polytechnic Institute , Worcester, MA	08/2019 - Present
Doctor of Philosophy in Bioinformatics and Computational Biology (Expected 12/2026)	
Santa Clara University , Santa Clara, CA	09/2014 - 06/2019
Master of Science in Bioengineering (Computational Biology track)	
Bachelor of Science in Bioengineering (Medical Device track)	

HONORS, AWARDS, & FELLOWSHIPS

2nd Poster Prize ISMB	07/2026
Poster Presentation in iRNA tract titled 'Proteogenomic machine learning for assessing the physiological fate of alternatively spliced protein isoforms'.	

Backlin Scholarship WPI	2026
Scholarship for last-year Ph.D. students	

R01LM014017 NIH	2021 - 2025
Research Assistant on 'Predicting the functional impact of alternative splicing on protein-protein interactions using an integrated approach' (PI: Prof. Dmitry Korkin)	

Julia Kasparian Fund for Neuroscience Research WPI	2019 – 2021
Research Assistant on 'Identify and monitor brain structure, function, and chemistry changes associated with dissociative disorders' (PI: Prof. Dmitry Korkin)	

RESEARCH EXPERIENCES

Worcester Polytechnic Institute , Worcester, MA	08/2019 - Present
Research Assistant, Advisor: Prof. Dmitry Korkin	
I'm a Ph.D. student in Korkin Lab working on a wide range of computational biology projects.	

- The Impact of Alternative Splicing on Human Proteome
 - Designed an integrative dataset from existing transcriptomic and proteomic datasets to imply protein isoform existence.
 - Developed robust, scalable computational pipelines to engineer features at RNA and protein level informed by both biological knowledge and LLMs.
 - Trained machine learning and deep learning models to reach over 80% accuracy.
 - Collaborated with Sheynkman Lab of University of Virginia to validate the model with disease related genes.
 - Applied the model to human genome to estimate the size human proteome.
- Large-Scale Comparative Genomics & Evolutionary Modeling
 - Systematically analyzed over 6,000 eukaryotic species of their exons and isoforms.
 - Proposed a model to explain the role of alternative splicing on the genome complexity of eukaryotes.
- Rapid SARS-Cov-2 Envelope and Protein Modeling
 - Built the first roadmap of SARS-Cov-2 structural genomics and interactomics using comparative modeling within a month into the pandemic.

- Collaborated with Marrink Lab of University of Groningen in Netherland to model the virus envelope structure and dynamics of SARS-Cov-2.
- Neuroinformatics & Machine Learning for Psychiatric Disease Prediction
 - Collaborated with McLean Hospital of Harvard Medical School to collect psychometric data and structural MRI data of female patients with trauma-related dissociation.
 - Built machine learning models to identify structural changes in suicidal patients' brains.

Beth Israel Deaconess Medical Center, Harvard Medical School, Boston, MA

07/2018 - 09/2018

Research Student, Advisor: Prof. Towia Libermann, MD

I worked as a summer research student with Prof. Libermann on metabolic diseases.

- Identifying proteins with predictive power on Pancreatic Ductal Adenocarcinoma and Nonalcoholic Steatohepatitis
 - Used data from ELISA and Somascan to identify protein candidates.
 - Preprocessed the data to overcome limited data size.
 - Developed machine learning models to predict related diseases with high accuracy.

Santa Clara University, Santa Clara, CA

09/2017 - 06/2019

Research Assistant, Advisor: Prof. Yuling Yan

I worked as a research assistant in Prof. Yan's lab for my bachelor and master thesis.

- Unsupervised Parkinson's Disease (PD) assessment
 - Proposed a low-cost and energy-efficient hybrid system that monitors patients' daily actions to quantify hand and finger tremors based on Unified Parkinson's Disease Rating Scale (UPDRS) tests.
 - Evaluated its characteristics in energy efficiency and medical accuracy and compared the device with existing products to highlight its salient feature.
 - Used different filters to analyze the EMG signal and give scores to patients wearing this device.
 - Explored multiple machine learning algorithms to classify a patient's actions into appropriate UPDRS tests.
- Multi-modality machine learning-based diagnosis for diseases
 - Used image, video, and audio recording of patients to train models to classify vocal fold diseases.
 - Used PET and fMRI scan and deep learning methods to classify multiple neurological diseases including Alzheimer's Disease, Parkinson's Disease, and seizure.
 - Converted ECG signals into a 2D representation and trained a CNN to detect Atrial Fibrillation.

The Institute of Medicinal Plant Development, Peking Union Medical College and Chinese Academy of Medical Sciences, Beijing, China

06/2016 - 09/2016

Research Student, Advisor: Prof. Xiaobo Sun

Xuanwu Hospital Capital Medical University, Beijing, China

06/2015 - 09/2015

Research Student, Advisor: Prof. Biao Chen, MD

RESEARCH TECHNIQUES & SKILLS

Programming languages: Python, Bash, MATLAB, Singularity/Apptainer/Docker, and SLURM (for High-Performance Computing).

Machine-learning related: training and finetuning biological large language models (LLMs), deep learning models (CNN, LSTM, RNN, etc.), supervised ML models (random forest, logistic regression, etc.), and reinforcement learning models.

Bioinformatics related: transcriptomic and proteomic data analysis and integration, protein homology and *ab initio* modeling, molecular dynamic simulation, MRI image processing.

MENTORSHIP AND TEACHING EXPERIENCES

Worcester Polytechnic Institute, Worcester, MA

08/2019 – present

- Mentoring MQP/IQP students
 - MQP/IQP is a year-long thesis project for junior and senior student. I mentor 5-10 students each year on deep learning analysis of neuroimages, leading to their contribution to published manuscripts.
- TA in CS 534 Artificial Intelligence
 - Graduate level class with ~70 students per semester. My responsibilities include hosting TA sessions every week to answer questions and grade homework and tests.

Santa Clara University, Santa Clara, CA

01/2019-04/2019

- TA in BIOE 162 BioSignals & Systems
 - Mandatory undergraduate level class for medical device track with ~ 40 students. My responsibilities involve leading 10 lab sessions on collecting and analyzing bio-signals including ECG, EEG, and EMG.

PUBLICATIONS, POSTERS, & PREPRINTS

Manuscripts in Preparation / Preprints:

Lu, Senbao, Yue Bao, Gloria M Sheynkman, and Dmitry Korkin. "Beyond Gene Length: Exon-Intron Architecture and Isoform Potential in the Evolution of Eukaryotic Complexity." *bioRxiv* (2026): 2026.03.25.714307. (In review *Genome Biology*)

Lu, Senbao, Ziyang Gao, Oleksandr Narykov, Gloria M Sheynkman, and Dmitry Korkin. "Proteogenomic machine learning for assessing the physiological fate of alternatively spliced protein isoforms." (In preparation for submission to *Nature Methods*)

Narykov, Oleksandr, Nathan T. Johnson, **Senbao Lu**, Shariq Madha-Krause, Yue Bao, Gloria Sheynkman, and Dmitry Korkin. "Quantifying functional impact of alternatively spliced isoforms."

Peer-reviewed publications:

Murali, Mayank, Jamie Saquing, **Senbao Lu**, Ziyang Gao, Emily F Watts, Ben Jordan, Zachary Peters Wakefield, Ana Fiszbein, David R Cooper, and Peter J Castaldi. "Biosurfer for Systematic Tracking of Regulatory Mechanisms Leading to Protein Isoform Diversity." *Genome Research* 35, no. 4 (2025): 1012–24.

Jogeshwar, Bhaavin K, **Senbao Lu**, and Benjamin C Nephew. "Neuroanatomical-Based Machine Learning Prediction of Alzheimer's Disease across Sex and Age." *Neuroscience* (2025).

Pezeshkian, Weria, Fabian Grünwald, Oleksandr Narykov, **Senbao Lu**, Valeria Arkhipova, Alexey Solodovnikov, Tsjerk A Wassenaar, Siewert J Marrink, and Dmitry Korkin. "Molecular Architecture and Dynamics of Sars-Cov-2 Envelope by Integrative Modeling." *Structure* 31, no. 4 (2023): 492–503. e7.

Wei, Ting-Ruen, **Senbao Lu**, and Yuling Yan. "Automated Atrial Fibrillation Detection with Ecg." *Bioengineering* 9, no. 10 (2022): 523.

Srinivasan, Suhas, Nathaniel G Harnett, Liang Zhang, M Kathryn Dahlgren, Junbong Jang, **Senbao Lu**, Benjamin C Nephew, Cori A Palermo, Xi Pan, and Mohamed Y Eltabakh. "Unravelling Psychiatric Heterogeneity and Predicting Suicide Attempts in Women with Trauma-Related Dissociation Using Artificial Intelligence." *European Journal of Psychotraumatology* 13, no. 2 (2022): 2143693.

- Polcari, Justin J, Ryan J Cali, Benjamin C Nephew, **Senbao Lu**, Mikhail Rashkovskii, Julianne Wu, Frances Saadeh, Eric Loucks, and Jean A King. "Effects of the Mindfulness-Based Blood Pressure Reduction (Mb-Bp) Program on Depression and Neural Structural Connectivity." *Journal of Affective Disorders* 311 (2022): 31–39.
- Gao, Ziyang, **Senbao Lu**, Oleksandr Narykov, Suhas Srinivasan, and Dmitry Korkin. "Structural Genomics and Interactomics of Sars-Cov2: Decoding Basic Building Blocks of the Coronavirus." In *Virus Bioinformatics*: Chapman and Hall/CRC, 2021.
- Srinivasan, Suhas, Hongzhu Cui, Ziyang Gao, Ming Liu, **Senbao Lu**, Winnie Mkandawire, Oleksandr Narykov, Mo Sun, and Dmitry Korkin. "Structural Genomics of Sars-Cov-2 Indicates Evolutionary Conserved Functional Regions of Viral Proteins." *Viruses* 12, no. 4 (2020): 360.
- Zhai, Yadong, Xiangbao Meng, Yun Luo, Yongmei Wu, Tianyuan Ye, Ping Zhou, Shilan Ding, Min Wang, **Senbao Lu**, and Lili Zhu. "Notoginsenoside R1 Ameliorates Diabetic Encephalopathy by Activating the Nrf2 Pathway and Inhibiting Nlrp3 Inflammasome Activation." *Oncotarget* 9, no. 10 (2018): 9344.
- Wang, Shan, Yu Tian, Jing-Yi Zhang, Hui-Bo Xu, Ping Zhou, Min Wang, **Sen-Bao Lu**, Yun Luo, Gui-Bo Sun, and Xu-Dong Xu. "Targets Fishing and Identification of Calendulose E as Hsp90ab1: Design, Synthesis, and Evaluation of Clickable Activity-Based Probe." *Frontiers in pharmacology* 9 (2018): 532.
- Adranly, Alexander, **Senbao Lu**, Yousef Zoumot, Yuling Yan, and Behnam Dezfouli. A Low-Cost and Energy-Efficient Platform for Unsupervised Parkinson's Disease Assessment. 2018 IEEE Global Humanitarian Technology Conference (GHTC): IEEE, 2018.

Book Chapters:

- Hosseini, Mohammad-Parsa, **Senbao Lu**, Kavin Kamaraj, Alexander Slowikowski, and Haygreev C Venkatesh. "Deep Learning Architectures." In *Deep Learning: Concepts and Architectures*: Springer International Publishing Cham, 2019.
- Yabar, AS Torres, R Cali, **S Lu**, A Howell-Munson, GJ Nephew, and BC Nephew. "The Impact of Ultrafine Particles on Mental Health."

Invited talks and posters:

- Lu S**, Gao Z, Narykov O, Sheynkman G, Korkin D. Proteogenomic machine learning for assessing the physiological fate of alternatively spliced protein isoforms. In: 34th Conference on Intelligent Systems for Molecular Biology (ISMB) iRNA Poster. ISCB; 2026 (2nd poster prize)
- Lu S**, Bao Y, Sheynkman G, Korkin D. Beyond Gene Length: Exon-Intron Architecture and Isoform Potential in the Evolution of Eukaryotic Complexity. In: 34th Conference on Intelligent Systems for Molecular Biology (ISMB) EvolCompGen Talk. ISCB; 2026
- Lu S**, Pezeshkian W, Grünwald F, Narykov O, Wassenaar TA, Marrink SJ, Korkin D. Structural characterization of M protein: Insights into a higher-order oligomerization in SARS-Cov-2 envelope and evolutionary relationship with ORF3a. In: 30th Conference on Intelligent Systems for Molecular Biology (ISMB) Poster Session. ISCB; 2022